

Headache and visual changes in a 19-year-old

What the first note left out, and what it changed

Case in brief

Kayla is a 19-year-old college student with about ten weeks of daily headache, a whooshing sound in her ears, and double vision on far left gaze. A first-year resident saw her earlier today and wrote a thorough note: a raised-pressure headache pattern, ibuprofen most days, a combined oral contraceptive, and normal strength, sensation, reflexes, and gait. Nobody had examined her eyes. You read that note, interviewed and examined her, and wrote your own assessment and plan. The intended diagnosis was idiopathic intracranial hypertension associated with doxycycline, with vision-threatening field loss, at a normal BMI.

What separated the diagnoses

SUPPORTING THE LEADING DIAGNOSIS

Doxycycline twice daily since July, for acne

Cycline antibiotics are reported in association with intracranial hypertension, and symptoms improved after stopping doxycycline in two reported cases. Large-cohort evidence is mixed, so call it an association (refs 3 to 5).

Bilateral papilledema on fundoscopy

Optic disc edema was the most predictive finding of true intracranial hypertension in one cohort, and the note never recorded the discs (ref 1).

Left abduction deficit, diplopia on left gaze

Matched the double vision already in the note and moved raised pressure from possible to likely.

Brief episodes of vision greying out

Transient visual obscurations, hidden by the note's "denies dizziness." They are strongly associated with papilledema (ref 7).

ARGUING AGAINST THE ALTERNATIVES

Papilledema and constricted fields with a normal focal neuro exam

Not a tension headache or migraine. The student-health label was an anchor, and misdiagnosis and bias in this presentation are well described (ref 1).

Ibuprofen 5 to 6 days a week for 5 weeks

Can sustain a daily headache. It does not explain the whooshing, the diplopia, or the papilledema.

SHOULD NOT BE MISSED

Raised-pressure picture with a combined oral contraceptive

Cerebral venous sinus thrombosis can look the same, and nothing in the history excludes it. Workup includes brain imaging with venography (ref 2), and thrombosis after lumbar puncture has been reported in this setting (ref 6).

Fields constricted worse on the left, acuity 20/20 and 20/25

Fields can be abnormal when acuity is near normal, so acuity alone would have reassured falsely (ref 7).

What we were given, and what was still needed

GIVEN

- Headache worse in the morning, lying flat, and with coughing, waking her from sleep; pulsatile whooshing; double vision on left gaze.
- Ibuprofen 5 to 6 days a week for 5 weeks; combined oral contraceptive; acne followed by dermatology.
- "Denies dizziness or lightheadedness," and an event at soccer practice with no detail.
- A neuro exam of strength, sensation, reflexes, and gait. No eye exam. A student-health label of tension headache.

TO TEST THE INHERITED DIAGNOSIS

A medication follow-up past the first answer

The first answer returned the birth control. A follow-up reached doxycycline, which the note's two-item list omitted.

Ask about vision, and what happened at practice

A dizziness question returns "No." A vision question returns brief episodes of greying out.

Ask about side vision or bumping into things

Reveals field loss by function: a mirror, a hallway, a ball, a cup.

TO FIND A DANGEROUS ALTERNATIVE

Fundoscopy

Bilateral papilledema. Without it the case stays a headache disorder (ref 1).

TO CHANGE MANAGEMENT

Eye movements and confrontation fields

A left abduction deficit and fields constricted worse on the left. These set the urgency of eye evaluation (ref 2).

Acuity and pupils

Near-normal acuity and no afferent pupillary defect. Worth recording, but least informative here.

Compare the differential

DIAGNOSIS	TYPICAL PATTERN AND CONTEXT	SUPPORTS IT HERE	ARGUES AGAINST OR DISTINGUISHES
* Idiopathic intracranial hypertension, doxycycline-associated	Positional and morning headache, whooshing, visual symptoms, papilledema, often a precipitant	Daily positional headache, whooshing, diplopia, papilledema, field loss, doxycycline	Imaging and lumbar puncture still needed. A normal BMI does not exclude it (ref 3)
Cerebral venous sinus thrombosis	Raised-pressure picture, with a hormonal or other prothrombotic context	Combined oral contraceptive, raised-pressure picture	Nothing in the history excludes it. Venography does
Mass lesion	Progressive headache, waking from sleep, focal signs	Daily progressive headache, waking from sleep, an aunt with a brain tumor	Normal focal exam. Imaging resolves it
Medication-overuse headache	Daily headache with frequent analgesic use	Ibuprofen 5 to 6 days a week for 5 weeks	Does not explain the whooshing, diplopia, or papilledema

* Intended diagnosis

Three take-home points

CARRY IT FORWARD *Before you accept a note, ask what its author never looked at and never asked.*

1. **A complete-looking note is a hypothesis, not a finding.** Here nobody had looked in her eyes, and the note's negatives answered a different question than the one that mattered. Read for what is absent as well as what is recorded.
2. **Ask the question that matches the symptom, then ask once more.** "Dizzy?" returned no and "Has your vision changed?" returned the answer. The first medication question returned the birth control and the follow-up returned doxycycline.
3. **In a headache with raised-pressure features, examine the eyes before you reassure.** Discs, eye movements, and fields set the urgency, and acuity can look normal while fields are not. When a dangerous diagnosis shares the picture, name the test that excludes it before the procedure.

References

1. Hansen NS, Korsbaek JJ, Hamann S, Jensen RH. Diagnostic process, misdiagnosis and bias in suspected idiopathic intracranial hypertension. *BMJ Neurol Open*. 2025;7(1):e000863. doi:10.1136/bmjno-2024-000863
2. Wakerley BR, Mollan SP, Sinclair AJ. Idiopathic intracranial hypertension: update on diagnosis and management. *Clin Med (Lond)*. 2020;20(4):384-388. doi:10.7861/clinmed.2020-0232
3. Ali NS, Long BD, Manzoor NF, Sismanis A, Coelho DH. Doxycycline-induced intracranial hypertension presenting as unilateral pulsatile tinnitus. *Otol Neurotol Open*. 2023;3(4):e043. doi:10.1097/ONO.0000000000000043
4. Angelette AL, Rando LL, Wadhwa RD, et al. Tetracycline-, doxycycline-, minocycline-induced pseudotumor cerebri and esophageal perforation. *Adv Ther*. 2023;40(4):1366-1378. doi:10.1007/s12325-023-02435-y
5. Eldweik L, McClelland C, Stein JD, Blachley TS, Lee MS. Association between cycline antibiotic and development of pseudotumor cerebri syndrome. *J Am Acad Dermatol*. 2019;81(2):456-462. doi:10.1016/j.jaad.2019.03.041
6. Garg A, Rajendram P, Muccilli A, Noel de Tilly L, Micieli JA. Dural venous sinus thrombosis after lumbar puncture in a patient with idiopathic intracranial hypertension. *Eur J Ophthalmol*. 2020. doi:10.1177/1120672120970406
7. Hayat SC, Yilmaz YC, Davulcu K, Ozal SA. Rising consultations for suspected papilledema: clinical and neuro-ophthalmologic insights. *Int Ophthalmol*. 2025;45(1):229. doi:10.1007/s10792-025-03603-5